Joaquin E. Rivas Sánchez

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Education

Universidad de las Ciencias Informaticas Feb 2022 — Jul 2025
Bachelor's in Computer Science Engineering (CSE) *Havana, CU*

- Cumulative GPA: 3.86/4.0 | Dean's List, Harvey S. Mudd Merit Scholarship, National Merit Scholarship
- Relevant Coursework: Machine Learning, Database Systems & Optimization, Data Structures, Linear Algebra, Discrete Mathematics, Multivariable & Single Variable Calculus

Moscow State University of Geodesy and Cartography (MIIGAiK) Oct 2024 — Jan 2025
Course - Modern methods and technologies of spatial data processing *Moscow, RU*

- Studied modern techniques for processing, analyzing, and managing spatial data using contemporary geodetic and GIS tools.
- Gained hands-on experience with advanced methods of spatial data acquisition, transformation, and accuracy evaluation.
- Explored current technologies in remote sensing, digital cartography, and geospatial data workflows.
- Learned practical approaches to handling large-scale geospatial datasets and integrating them into mapping and analysis systems.

Work Experience

Teaching Assistant - Machine Learning Sep 2025 — Present
Universidad de las Ciencias Informaticas *Havana, CU*

- Delivered hands-on practical sessions covering Machine Learning pipeline, Python, NumPy, pandas, Scikit-Learn.
- Assisted the main professor during lecture classes and supported course planning and material preparation.
- Graded assignments, exams, and machine-learning projects fairly and accurately.
- Provided one-on-one tutoring and academic support to students.
- Helped improve course exercises and contributed to updating the curriculum.

Projects

RAG Chatbot (github.com/humankernel/rag-revamped) Jan 2025 — Jun 2025

- Built a complete RAG pipeline (chunking, embeddings, vector search, context construction, LLM generation).
- Implemented modular components: loader, chunker, embedder, retriever, aggregator, and generator.
- Integrated with OpenAI and vLLM models for flexible local or API-based inference.
- Created evaluation tools for retrieval quality, latency, and model output accuracy.
- Wrote full system documentation with architecture diagrams and unit tests.

Chess Engine (github.com/humankernel/chess-engine) Aug 2025 — Oct 2025

- Implemented a full bitboard-based chess engine in C using 64-bit board representation.
- Built attack tables for pawns, knights, and kings, including file masks and edge conditions.
- Designed evaluation heuristics: material, pawn structure (isolated, doubled, blocked), and mobility.
- Implemented utility functions for bit manipulation: LSB pop, bit counting, and file scanning.
- Built the foundation for a move generator and minimax search using efficient attack masks and occupancy handling.

HTTP Server from Scratch (github.com/humankernel/http-server) Apr 2025 — Apr 2025

- Built a minimal HTTP/1.1 server in Go from the ground up using sockets without external frameworks.
- Implemented request parsing, routing, concurrency handling, and static file serving.
- Added proper response headers, MIME type detection, and error handling.
- Gained deep understanding of TCP networking, low-level protocols, and server architecture.

Terminal Chat (github.com/humankernel/chat-go) Apr 2025 — Apr 2025

- Built a real-time chat system in Go using WebSockets, with both client and server implemented from scratch.
- Designed a custom chat protocol supporting normal messages, room changes, user presence, and file transfer.
- Implemented a multi-room architecture with a central Hub managing user registration, broadcasts, and concurrency using goroutines and channels.
- Added file transfer capability over WebSockets, including metadata negotiation, binary transfer, and per-user routing.

- Used synchronization primitives (sync.Mutex, channels) to safely handle concurrent users and avoid race conditions.
- Implemented a terminal client with username login, message handling, and file sending using raw I/O and WebSocket frames.

Skills

- **Programming Languages**: C, Python, Typescript, Go, HTML/CSS, SQL (PostgreSQL)
- **Technologies**: Linux, Git, Docker, React, Tanstack's Stack, Astro, NextJS, Tailwind CSS, Pandas, NumPy
- **Human Languages**: Spanish (native), English (B1)